REFONTE INTERNATIONAL TRAINING AND INTERNSHIP PROGRAM

Astrodynamics Specialist

SESSION 5

Satellite Constellations and Coverage Analysis

Footprints, Ground Tracks, Walker Patterns, and Designing for Global Coverage

Prepared for Cao Khanh Tan

Module 5, opening the Advanced Application Phase

Builds on Sessions 1 to 4: orbital elements, inclination, period, transfers

Project Design and compare two LEO constellations for global coverage

Tools STK Coverage Definition and Figure of Merit, Walker tool, Python

SESSION 5

Satellite Constellations and Coverage Analysis

One satellite sees only a patch of the Earth. To watch a whole region continuously, or to give every point a navigation or internet signal, you arrange many satellites into a constellation. This is the mission-design heart of the course. We work out how much one satellite sees, how long it stays in view, how its ground track sweeps the surface, and how Walker patterns tile the globe. Then we put it to work: the assignment is a real design project, sizing and comparing two LEO constellations for global coverage and deciding which balances satellite count, revisit and performance best.

Learning objectives

- Relate altitude and minimum elevation angle to footprint, coverage fraction and access duration.
- Compute a ground track and find when it repeats.
- Define revisit time and coverage figure of merit, and reason about global versus regional coverage.
- Read Walker Delta and Walker Star notation and compute plane and phase spacing.
- Trade altitude, inclination, satellite count and revisit against each other.
- Design, simulate and compare two constellation architectures and justify a choice.

> **The design question**
> Higher orbits see more of the Earth each, so you need fewer satellites, but pay in launch delta-v and signal latency. Lower orbits see less each, so you need many more, but they are cheap to reach and quick to talk to. Every decision in this module is a point on that trade.

PART 1

Coverage Geometry

A ground station can only use a satellite above a minimum elevation angle epsilon, set by terrain and atmosphere. That single constraint fixes the geometry: one triangle joining the Earth centre, the satellite, and the ground point at the edge of coverage. The nadir angle follows from the law of sines, the Earth central angle is what is left, and the slant range closes the triangle:

$$\sin\eta = \frac{R_E}{R_E + h}\cos\varepsilon$$

$$\lambda = 90^\circ - \varepsilon - \eta$$

$$\rho = (R_E + h)\,\frac{\sin\lambda}{\cos\varepsilon}$$

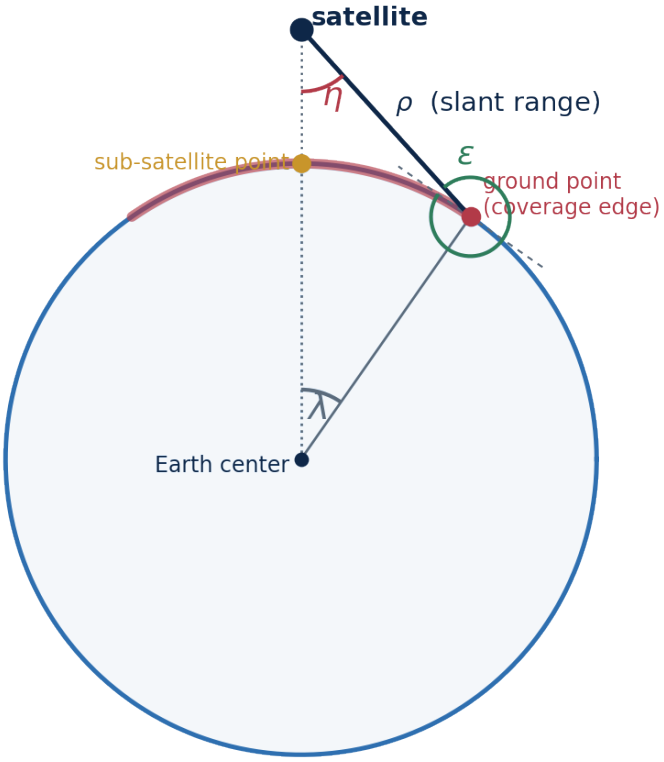


Figure 1. The coverage triangle. The nadir angle eta is at the satellite, the elevation epsilon at the ground point, and the Earth central angle lambda at the centre; they sum with a right angle. The slant range rho is the true line of sight, and it sets the signal latency.

PART 2

Footprint, Access and How Much One Satellite Sees

The satellite covers a spherical cap of half-angle lambda around its sub-satellite point. Two numbers summarise it, the footprint radius on the ground and the fraction of the Earth inside it:

$$d = R_E \lambda$$

$$f = \frac{1 - \cos\lambda}{2}$$

Both grow quickly with altitude (Figure 2): lambda climbs from a few degrees in low orbit to over 70 degrees at MEO. A satellite does not sit still over a point, though, so a second quantity matters, the access duration, the time a satellite stays above the elevation mask during a pass. For an overhead pass it is roughly the fraction of the orbit spent crossing the coverage cap:

$$t_{\text{acc, max}} \approx \frac{\lambda}{180^\circ}\, T$$

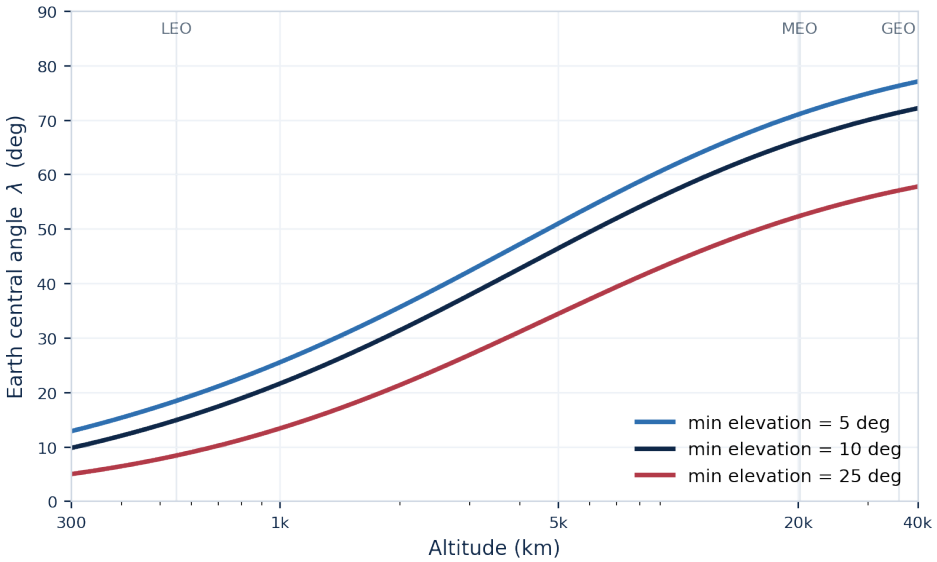


Figure 2. Earth central angle versus altitude for three elevation masks. A stricter mask (25 degrees) shrinks coverage; a relaxed one (5 degrees) extends it toward the horizon.

As an absolute lower bound, tiling the sphere with static caps needs at least N caps; a moving constellation needs more to keep something always in view:

$$N \geq \frac{2}{1-\cos\lambda}$$

Worked example: 550 km versus GPS
At 550 km with a 25 degree mask, lambda is only 8.5 degrees: the footprint radius is 941 km, one satellite sees 0.5 percent of the Earth, and an overhead pass lasts about 4.5 minutes. A GPS satellite at 20,200 km with a 5 degree mask has lambda = 71 degrees, sees 34 percent of the Earth at once, and stays in view for over an hour. The low satellite must be one of very many; the high one can be one of a few.

PART 3

Ground Tracks

A ground track is the sub-satellite path on the rotating Earth. Because the planet turns eastward under the orbit, each pass crosses the equator farther west, by the fraction of a day the orbit takes:

$$\Delta L = -360^\circ \frac{T}{T_\oplus}$$

The maximum latitude a track reaches equals the inclination (Session 1), so a constellation that must cover the poles needs near-polar planes. A track repeats when the satellite completes whole revolutions in whole days:

$$k\,T = m\,T_\oplus \quad (k \text{ revs},\ m \text{ days})$$

Worked example: two very different tracks

GPS has an 11.97 hour period, almost exactly half a sidereal day, so it makes two revolutions per day and its ground track repeats daily. A 550 km satellite has a 1.59 hour period, about 15 revolutions per day, and its track shifts roughly 24 degrees west each pass, sweeping the whole planet in longitude over a day.

PART 4

Revisit Time and Figure of Merit

Coverage is judged by a small set of metrics that a tool like STK computes over a grid of target points and reports as a figure of merit (FOM):

- **Percent coverage:** the share of the target region in view at a given instant, or averaged over time.
- **Revisit (or gap) time:** how long a point waits between successive accesses. Mean and maximum gap both matter.
- **Access duration:** how long each individual pass lasts, which drives data volume and contact time.
- **Fold of coverage:** how many satellites see a point at once; navigation needs at least four for a fix.

For continuous global coverage the maximum revisit gap should fall to zero everywhere. When it does not, the size of the worst gap tells you where the constellation is too thin. These are exactly the numbers we will compute for the design project, and they are what STK Coverage Definitions and Figure-of-Merit objects report as colour maps over the Earth.

PART 5

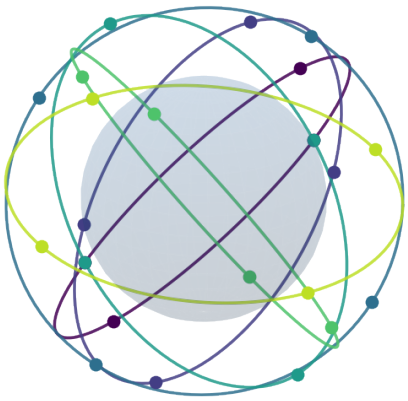
Constellation Patterns: Delta and Star

Two families dominate, and both are captured by the Walker notation, which packs the whole geometry into four numbers with i the inclination, T the total satellites, P the equally spaced planes, and F an integer that phases adjacent planes:

$$\text{Walker } i : T/P/F$$

$$\Delta\Omega = \frac{360^\circ}{P}, \quad \Delta u = \frac{360^\circ}{T/P}, \quad \Delta\phi = F\,\frac{360^\circ}{T}$$

- **Walker Delta:** the P planes are spread over a full 360 degrees of RAAN. This is the standard for near-global coverage and is what GPS (55 deg : 24/6/2) and Galileo (56 deg : 24/3/1) use.
- **Walker Star:** the planes are spread over 180 degrees and are near-polar, so ascending and descending passes interleave and the poles are covered. Iridium (86.4 deg, 6 planes, 66 satellites) is the classic streets-of-coverage Star.



Walker 55 deg : 24/6/2 (6 planes, 4 satellites each)

Figure 3. A Walker Delta constellation, 55 degrees with 24 satellites in 6 planes. Planes are evenly spaced in RAAN and satellites are phased between planes, spreading coverage evenly. A Star pattern instead clusters the planes near the poles.

Global versus regional

A Delta at moderate inclination gives dense coverage in a latitude band and leaves the poles thin; a Star gives true pole-to-pole global coverage. If a mission only needs one region, a small set of well-phased planes (or even a few highly eccentric Molniya orbits) beats a full global constellation. Match the pattern to the coverage requirement, not the other way round.

PART 6

Design Project: Two LEO Constellations Compared

To make the trade concrete, here are two candidate LEO constellations, both aiming at near-global coverage with a 10 degree elevation mask, simulated over 24 hours on a global grid. They differ in the single most important lever, altitude, and everything else follows from it.

Metric	A: low shell	B: mid shell
Altitude / inclination	550 km / 80 deg	1,200 km / 87 deg
Walker pattern	120 / 10 / 1	48 / 8 / 2
Per-sat central angle	15.0 deg	24.0 deg
Coverage (24 h, area-weighted)	92.5 percent	93.1 percent
Mean revisit gap	6.6 min	3.1 min
Worst gap	28 min	12 min
Access per pass (max)	7.9 min	14.6 min
Edge latency (one-way)	6.1 ms	10.4 ms
Satellites required	120	48

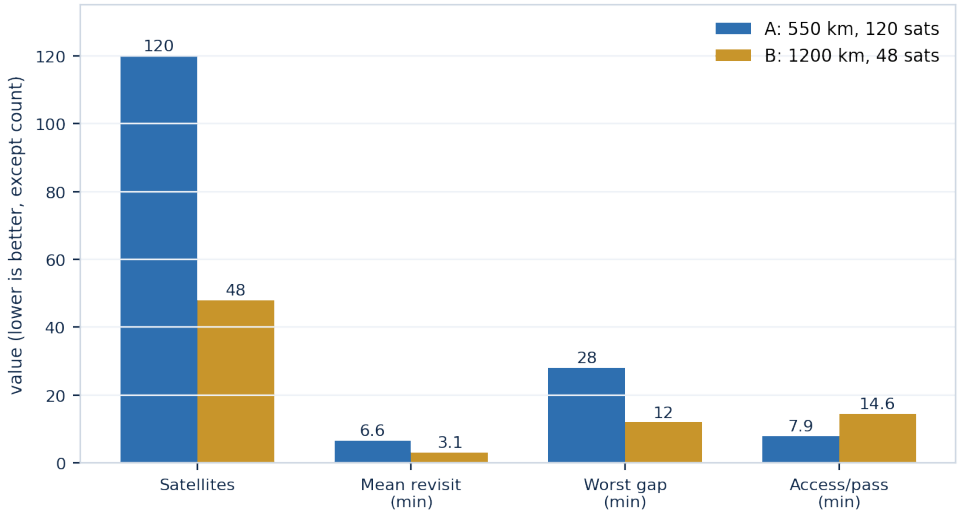


Figure 4. The two designs head to head. B reaches the same coverage as A with better revisit using fewer than half the satellites, because footprint area grows steeply with altitude. A trades that away for lower latency and shorter, more frequent passes.

Which is the better balance?

B is the more efficient global design: equal coverage, roughly half the revisit, and 40 percent of the satellites. Its costs are higher latency, more launch delta-v per satellite, and a longer access that is actually helpful for communications dwell. A wins only where low latency is critical or where mass-produced satellites make a large count cheap, which is exactly the reasoning behind low mega-constellations like Starlink. For a general global mission that does not need millisecond latency, B is the better balance of count against revisit.

Computing Coverage, and the STK Workflow

The per-satellite geometry is a handful of lines; the constellation statistics come from propagating every satellite and testing each grid point against the coverage cap over time. Figure 5 is the resulting figure-of-merit map for design B.

```
import numpy as np
Re, mu = 6378.137, 398600.4418
def coverage(h, eps_deg):
    r=Re+h; eps=np.radians(eps_deg)
    eta=np.arcsin(Re/r*np.cos(eps)); lam=np.pi/2-eps-eta
    return np.degrees(lam), Re*lam, (1-np.cos(lam))/2   # deg, km, fraction

# A grid point is 'covered' at time t if ANY satellite's sub-point
# lies within the central angle lam of it. Propagate the Walker set,
# accumulate covered/not over 24 h, then reduce to percent coverage,
# mean and worst revisit gap per point -> the figure of merit.
```

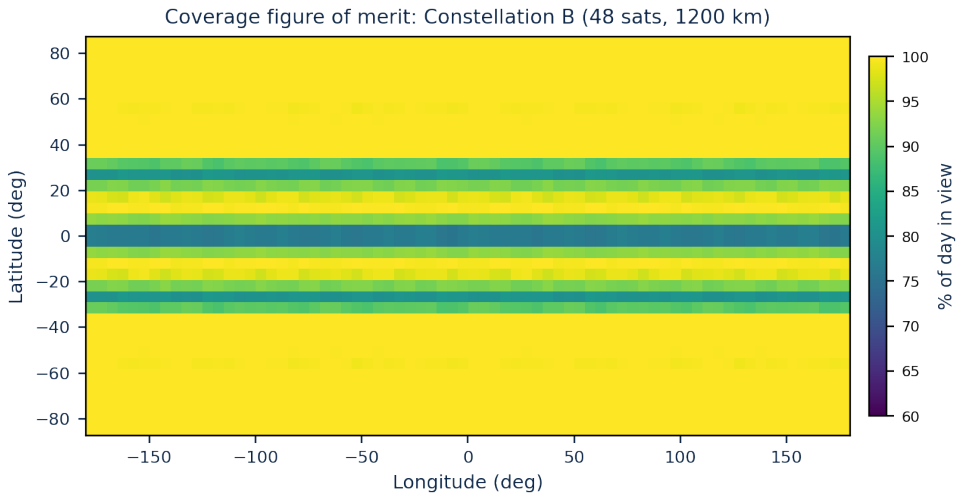


Figure 5. Figure-of-merit map for design B: percent of the day each point is in view. The near-polar Star pattern covers high latitudes almost continuously and thins toward the equator, the mirror image of a low-inclination Delta.

The STK workflow

In STK you build each design with the Walker tool (enter i : T/P/F directly), define a Coverage Definition over a latitude-longitude grid, and attach a Figure of Merit set to Simple Coverage, Revisit Time or Access Duration. STK then colours the globe exactly like Figure 5 and reports the mean and maximum per point. GMAT and a short Python script reproduce the same numbers; the hand geometry tells you whether to trust them.

PRACTICE

Design Project (aim for 12 to 15 hours)

This week is a single project: design and compare two LEO constellations for global Earth coverage, then argue which gives the best balance of satellite count, revisit and coverage. Bring theory, calculations, STK screenshots and a written justification.

Assignment: Design and Compare Two LEO Constellations

- **Theory (2 to 3 h).** In your own words, explain how altitude, inclination, elevation mask and satellite count each affect coverage and revisit. State your coverage requirement (for example: continuous coverage of all latitudes below 70 degrees, or a maximum revisit gap under 15 minutes).
- **Design two architectures (3 to 4 h).** Choose two different LEO designs, for example a low high-count shell and a higher lower-count shell. For each, compute the central angle, footprint, and maximum access duration by hand, and write the Walker notation i : T/P/F with its plane and phase spacing.
- **Size and simulate (3 to 4 h).** In Python or MATLAB, estimate percent coverage and mean and worst revisit for each design over 24 hours (the loop in the notes above). Report the numbers in a table like the one in Part 6.
- **STK (3 to 4 h).** Build both constellations with the STK Walker tool, add a Coverage Definition and a Figure of Merit for revisit time, and capture the FOM map and statistics for each. Screenshots required.
- **Justify (1 to 2 h).** Recommend one design for your stated requirement, with a short justification that weighs satellite count against revisit, latency and launch cost.

Reference results (the two designs in Part 6)

Quantity	Design A (550 km)	Design B (1200 km)
Central angle lambda (10 deg mask)	15.0 deg	24.0 deg
Footprint radius	~1,670 km	~2,670 km
Coverage over 24 h	92.5 percent	93.1 percent
Mean / worst revisit	6.6 / 28 min	3.1 / 12 min
Satellites	120	48

Your own two designs will differ; these are the yardstick to check your simulation against for the same altitudes and masks.

Coming up in Session 6

Rendezvous and proximity operations: relative motion in orbit, the Clohessy-Wiltshire equations, and how one spacecraft approaches and docks with another. The last building block before the capstone.

REFERENCE

One-Page Formula Sheet

Nadir angle from elevation	$\sin\eta = \frac{R_E}{R_E + h}\cos\varepsilon$
Earth central angle	$\lambda = 90^\circ - \varepsilon - \eta$
Slant range	$\rho = (R_E + h)\frac{\sin\lambda}{\cos\varepsilon}$
Footprint radius	$d = R_E\lambda$
Coverage fraction	$f = \frac{1-\cos\lambda}{2}$
Max access per pass	$t_{\text{acc, max}} \approx \frac{\lambda}{180^\circ}T$
Ground-track shift per rev	$\Delta L = -360^\circ\frac{T}{T_\oplus}$
Repeating ground track	$kT = mT_\oplus \quad (k \text{ revs},\ m \text{ days})$
Walker notation	$\text{Walker}\ \ i: T/P/F$
Walker plane and phase spacing	$\Delta\Omega = \frac{360^\circ}{P}, \quad \Delta u = \frac{360^\circ}{T/P}, \quad \Delta\phi = F\frac{360^\circ}{T}$
Static minimum satellites	$N \geq \frac{2}{1-\cos\lambda}$

Key glossary

- **Elevation mask:** the minimum angle above the horizon for a usable satellite; a stricter mask shrinks coverage.
- **Earth central angle (lambda):** the half-angle of the coverage cap, seen from the Earth centre.
- **Access duration:** how long a single pass stays in view; sets contact and data-downlink time.
- **Revisit time:** the gap a point waits between passes; the key figure of merit for coverage.
- **Walker Delta / Star:** planes spread over 360 degrees (near-global) or 180 degrees near-polar (pole-to-pole).

Going further

For depth, see Wertz and Larson, Space Mission Analysis and Design (coverage and constellation chapters), Vallado (ground tracks and access), and Curtis (constellation geometry). These are real, widely used texts; please confirm editions yourself, as I cannot verify those details from here.